Genomic diversity of pussy willow (*Salix discolor*) across the Northeastern United States

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Background

Salix discolor (pussy willow) is a dioecious shrub willow native to North America.

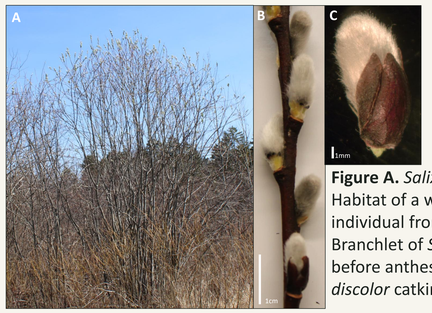


Figure A. *Salix discolor* profile. (A) Habitat of a wild *Salix discolor* individual from Portland, ME. (B) Branchlet of *S. discolor* with catkins before anthesis. (C) Emerging *S. discolor* catkin from floral bud.

This species blooms in the late winter around mid-March providing nectar to early emerging pollinators such as the Frigid mining bee (*Andrena frigida*) and the Unequal cellophane bee (*Collete inaequalis).*



Little is known about how genetic diversity in *S. discolor* is distributed throughout the Northeastern United States. We used whole-genome sequencing to measure measure genetic diversity and population structure to place *S. discolor* among co-occurring willows across EPA Level III Ecoregions 58, 59 and 82.

Knowing species identity and diversity helps guide conservation efforts to preserve native populations and support the selection of locally adapted plant material for restoration or landscape improvement activities.

Questions

1. Do genetic clusters match field identifications of *S. discolor* and co-flowering willows, and are any samples hybrids?
2. How much genetic diversity does *S. discolor* hold, and is it structured by ecoregion?
3. Is genetic variation associated with environment, suggesting local adaptation? *(in progress)*

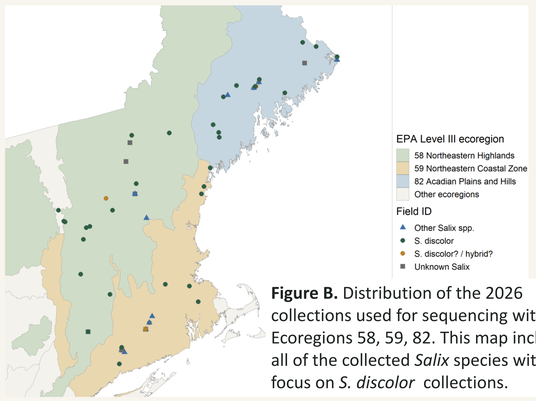


Figure B. Distribution of the 2026 collections used for sequencing within Ecoregions 58, 59, 82. This map includes all of the collected *Salix* species with a focus on *S. discolor* collections.

Sequencing & pipeline

Samples

94 Willow individuals sequenced

~53 *S. discolor*

Remaining samples: other *Salix* species, identified in the field including *S. sericea, S. petiolaris, S. humulis, S. atrocinerea, S. cinerea, S. caprea*

Ultima Genomics UG100, single-end ~200 bp reads, ~50 M reads per sample (≈20-30× coverage).

1 · Align
BWA-MEM to the *S. purpurea* 94006 v5.1 reference (Phytozome 519; assembly v5.0)

↓

2 · Mark duplicates
Picard, for single-end reads

↓

3 · Call variants
bcftools 1.23 mpileup + multiallelic caller with all 94 samples

↓

4 · Filter
Genotype depth ≥10, ≤20% missing, QUAL ≥30, site-depth cap; biallelic SNPs, MAF ≥5%

↓

5 · Population genomics
Thinned SNPs → PCA, ancestry (ADMIXTURE model), heterozygosity, pairwise relatedness

↓

6 · Environment association (in progress)
Site climate variables → RDA and LFMM, correcting for population structure

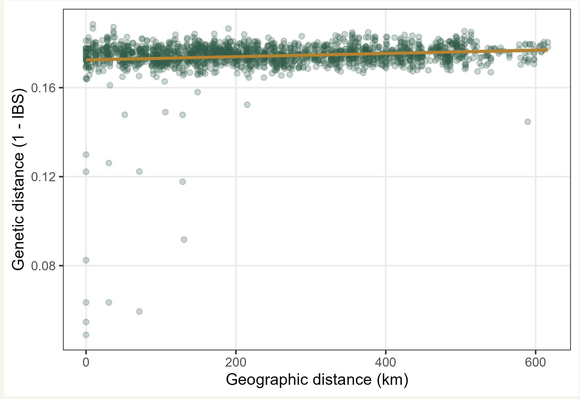


Figure F. Weak isolation by distance across 600 km
Genetic vs geographic distance. Mantel r = 0.138, p ≤ 0.001.

Results

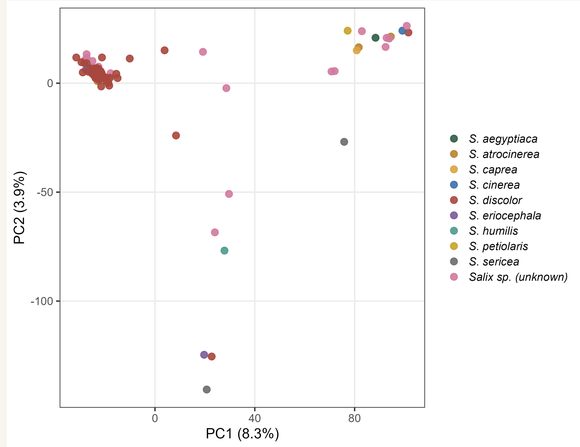


Figure B. WGS resolves willow species and corrects field IDs PCA of 209,879 genome-wide SNPs (90 plants). PC1 separates *S. discolor* from other *Salix* species confirming 47 of 53 field-identified and 16 of 37 uncertain, unknown or other-species plants were genetically *S. discolor*.

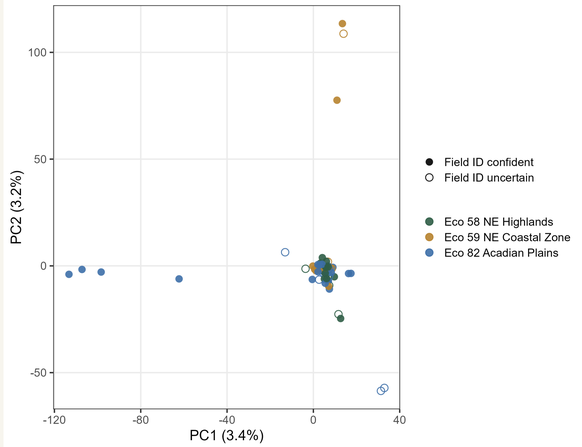


Figure C. No genetic structure among ecoregions (FST ≈ 0)
PCA of 53 unrelated, genetically confirmed *S. discolor*. Ecoregions overlap completely with a pairwise FST of -0.004 to -0.002.

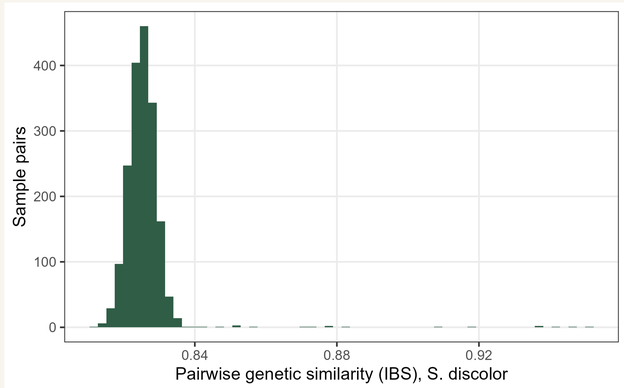


Figure D. Histogram of every *S. discolor* individual
Pairwise genetic similarity (IBS) among 1,830 *S. discolor* pairs.

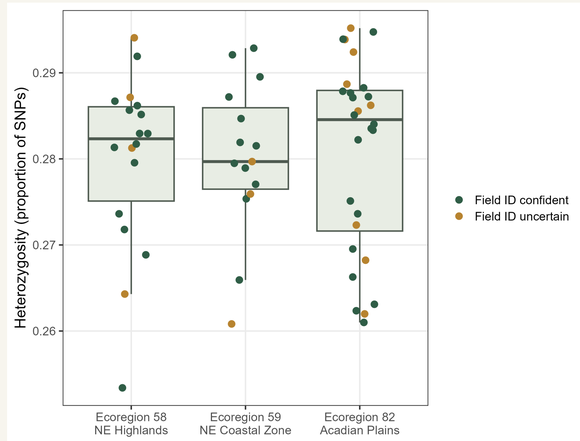


Figure E. Genetic diversity is uniform across the region
Observed heterozygosity of 61 *S. discolor* by ecoregion (medians 0.282, 0.280, 0.285; Kruskal-Wallis p = 0.70).

Implications for practitioners

- Ecoregions 58, 59 and 82 can be treated as one genetic seed zone with even diversity across the Northeast.
- Restoration activities should use cuttings from many unrelated donors across the landscape from both sexes to maintain high genetic diversity found in wild populations.
- **If possible, confirm species identity of donor plants, via genomics, before propagation!**

Next steps

- Genome-wide environmental association (climate, soil, latitude) on the full 45.5 M variant set with indels and SNPs
- Link common-garden phenology, eg. flowering time, to genotype and source latitude
- Validate calls with DeepVariant (Ultima's model) vs. BCFtools
- Sex identification from the sex-determining region to test sex ratios
- Additional surveys for non-native *Salix* species

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Questions?

Other Updates →